

## DES FILE TRANSFER

### CLIENT TO SERVER

This file is used for demonstrating secure file transfer using the Data Encryption Standard (DES) algorithm in a Python client-server application.

The client reads the file as binary data and encrypts the data using DES before sending it to the server.

The encrypted data is called ciphertext. The server receives the ciphertext and decrypts it using the same DES key to recover the original plaintext.

Algorithm: Data Encryption Standard (DES)

Communication: TCP Client-Server

Direction: Client to Server

Encryption: Client Side

Decryption: Server Side

File Type: PDF

Required File Size: 1 MB

The file transfer process is:

Original File



Read Binary Data



DES Encryption



Ciphertext



TCP Socket



Server



DES Decryption



Original Plaintext File

Purpose:

The purpose of this experiment is to demonstrate encryption, transmission, and decryption of a binary file using DES. The server should recover the same original file after decryption.

This experiment is part of the Active Learning Management (ALM) activity for Cryptography and Network Security.

Client: Sends the encrypted file.

Server: Receives and decrypts the encrypted file.

The plaintext and ciphertext sizes are displayed at both sides to verify the file transfer.

End of File.